Telecommunication Assignment - 5

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What is antenna?

An antenna is a metallic structure that captures and/or transmits radio electromagnetic waves. Antennas come in all shapes and sizes from little ones that can be found on our roof to watch TV to really big ones that capture signals from satellites millions of miles away.

The antennas that Space Communications and Navigation (SCaN) uses are a special bowl shaped antenna that focuses signals at a single point called a parabolic antenna. The bowl shape is what allows the antennas to both capture and transmit electromagnetic waves. These antennas move horizontally (measured in hour angle/declination) and vertically (measured in azimuth/elevation) in order to capture and transmit the signal. Fig 1.1 is a 70 meter antenna that found outside of Madrid, Spain and which can capture and transmit signals to the furthest depths of the solar system.

Fig 1.1 : Example of the parabolic antenna

Both co-linear antennas and antenna arrays provide enhanced performance compared to single antennas. Co-linear antennas offer improved gain and radiation pattern focusing in a specific direction, while antenna arrays enable beam forming, increased gain, and beam steering capabilities. The specific design and configuration of these antennas can vary depending on the application and desired performance parameters.

Types of antenna :

Antenna	Summary	Gain	Radiation Pattern	Bandwidth	Impedance	Size
Dipole Antenna	Consists of two conductive elements aligned along the same axis	Moderate	Omnidirectional	Wide	Balanced	Moderate
Yagi-Uda Antenna	Comprises a driven element, one or more directors, and a reflector, arranged in a linear array	High	Highly Directional	Moderate	Balanced	Moderate to larger size
Patch Antenna	Flat, Planar antenna with a radiating patch on a dielectric substrate	Low to moderate	Typically Directional	Wide	Can Vary according to configuration	Compact
Parabolic Reflector Antenna	Employs a parabolic-shaped reflector to focus incoming waves onto a smaller feed antenna	High	Highly Directional	Narrow	Can Vary according to configuration	Large
Co-Linear Antenna	Co-Linear antennas consist of multiple antennas aligned along the same axis, typically in a vertical configuration.	High	Depends on the design of antenna	Wide	Balanced	Moderate
Antenna Arrays	An antenna array is a collection of multiple individual antennas arranged in a specific pattern or configuration	High	Directional	Wide	Can Vary according to configuration	Compact to larger size

Table 1.1 : Comparison of different types of antennas

References

[1] Thuy Mai 'What is an antenna?' [online], August 7th 2017, https://www.nasa.gov/directorates/heo/scan/communications/outreach/funfacts/txt_antenna.html [accessed 24.05.2023]

[2] ANSY BLOG 'An Introduction to Common Antenna Designs' [online], February 21st 2023, https://www.ansys.com/blog/common-antenna-designs [accessed 24.05.2023]

[3] 'Antenna Theory - Types of Antenna' [online], https://www.tutorialspoint.com/antenna_theory/antenna_theory_types_of_antennas.htm [accessed 24.05.2023]